

Computer Graphics – October 12th, 2023

Name: _____

Matriculation Number: _____

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (18 pages on 9 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If you need additional space, you can find empty pages at the end of the exam
- If needed, we staple additional blank sheets.
- No other material is allowed!

I have read and understood the instructions above.

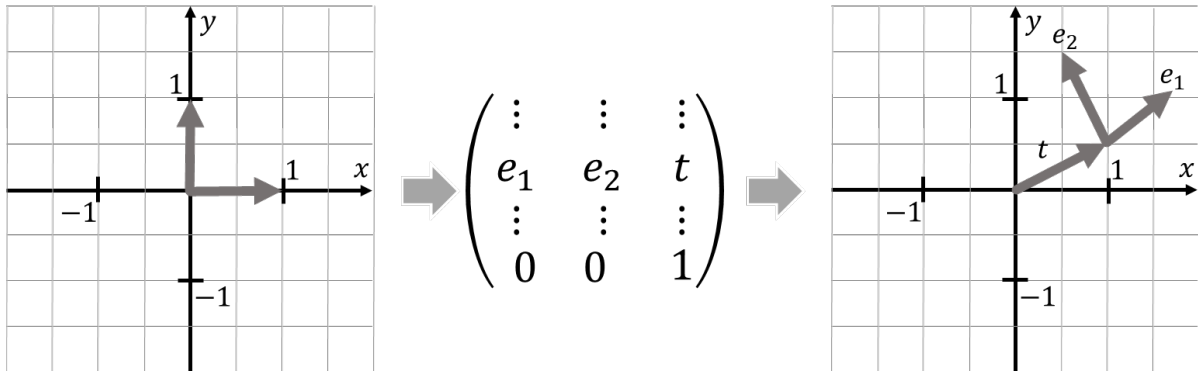
Erlangen, October 12th, 2023 _____
Signature

1	2	3	4	5	6	ALL
13	10	10	9	10	8	60

Grade:

Assignment 1: Affine Mappings in 2D [13 points]

Affine mappings in **2D** can be expressed using **homogeneous coordinates** and 3×3 matrices with the following structure, where $(\vec{e}_1, \vec{e}_2)$ is the linear part (image of base vectors) and $\vec{t}$ is the translation:



Specify such matrices for the following mappings of 2D points:

a) **[1P]** Uniform scaling by a factor of 2

$$M = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

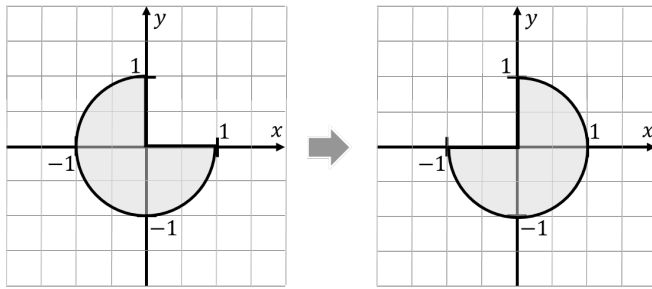
b) **[1P]** Mapping from meters to millimeters

$$M = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

c) **[1P]** Rotation by 90 degrees (counter clockwise)

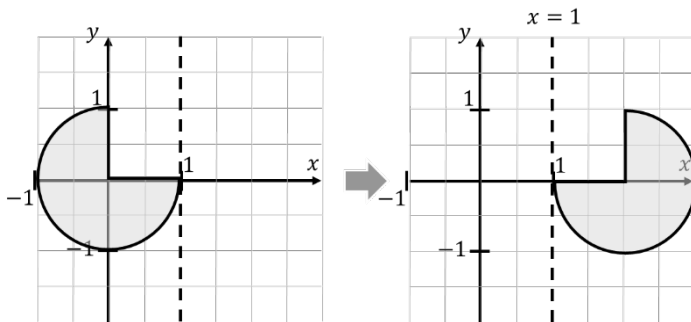
$$M = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

d) **[1P]** Reflection at y -axis



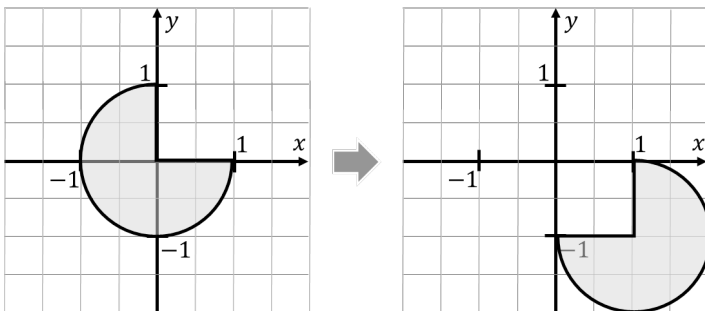
$$M = \begin{pmatrix} & \\ & \end{pmatrix}$$

e) **[2P]** Reflection at line $x = 1$



$$M = \begin{pmatrix} & \\ & \end{pmatrix}$$

f) **[2P]** a rotation and a translation



$$M = \begin{pmatrix} & \\ & \end{pmatrix}$$

g) **[3P]** Which properties does the mapping described by the following matrices fulfil?

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{is} \quad \begin{array}{l} \square \text{ affine} \quad \square \text{ a shearing} \quad \square \text{ a translation} \end{array}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \quad \text{is} \quad \begin{array}{l} \square \text{ projective} \quad \square \text{ a reflection} \quad \square \text{ a translation} \end{array}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{is} \quad \begin{array}{l} \square \text{ affine} \quad \square \text{ linear} \quad \square \text{ a translation} \end{array}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \quad \text{preserves} \quad \begin{array}{l} \square \text{ size} \quad \square \text{ angles} \quad \square \text{ ratios} \end{array}$$

h) **[2P]** Describe the following projective mapping by a homogeneous 4x4 matrix:

$$(x, y, z) \rightarrow \left(\frac{x}{y+1}, \frac{x+3}{y+1}, \frac{x+z}{y+1} \right)$$

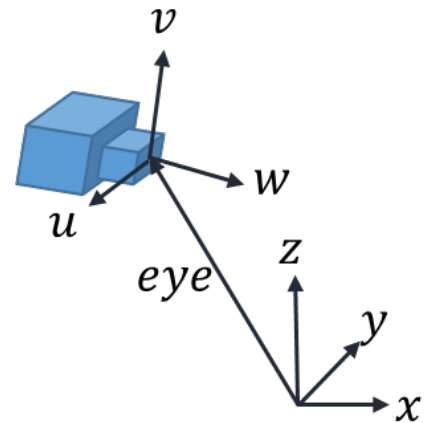
$$M = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

Hint: First write down the mapping in homogeneous coordinates.

Assignment 2: Viewing and Perspective [10 points]

a) **[4P]** Specify the camera coordinate system for a camera at position $(1,2,6)$ looking at a point $(1,2,3)$ with up-vector $(0,1,0)$.

Use the convention from the picture on the right, with w pointing in positive gaze direction. Don't care about right- or left-handed coordinate systems. Note: that image on the right does not show this particular setup!



$$eye = \left(\begin{array}{|c|} \hline \\ \hline \end{array} \right)$$

$$u = \left(\begin{array}{|c|} \hline \\ \hline \end{array} \right)$$

$$v = \left(\begin{array}{|c|} \hline \\ \hline \end{array} \right)$$

$$w = \left(\begin{array}{|c|} \hline \\ \hline \end{array} \right)$$

b) **[1P]** The above computed vectors define the camera matrix, which is a rotation and a translation.

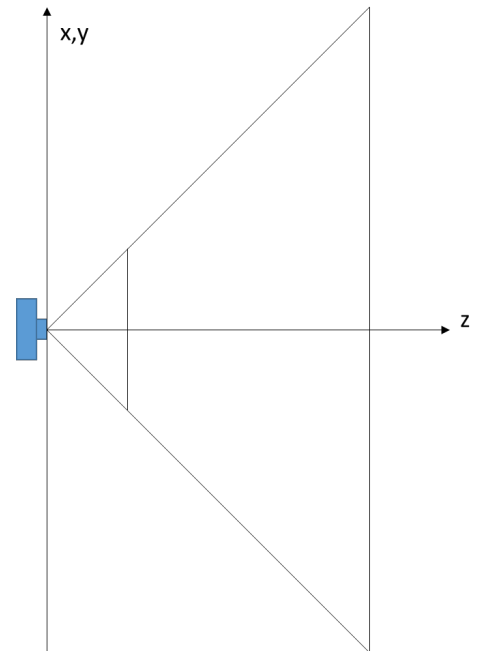
If we want to animate the camera, it is not a good idea to interpolate these matrices. Which other rotation representation is better suited for interpolation?

c) **[2P]** For normalization (perspective projection) we use a viewing frustum as shown on the right. The window on the near plane is defined by $x_{min} = y_{min} = -1$, $x_{max} = y_{max} = 1$, near and far plane are at $near = 1$, $far = 1001$.

What is the field of view and the aspect ratio of this perspective?

field of view =

aspect ratio =



d) **[2P]** How does the projection matrix look in this case (see relevant lecture slide at the bottom)

$$P = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

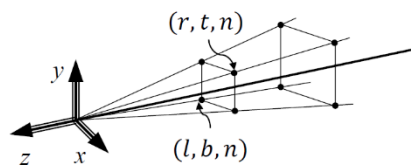
e) **[1P]** In which WebGL shader (vertex or fragment) is the projection matrix P always needed?

WebGL: Perspective Projection Matrix



• Perspective Projection:

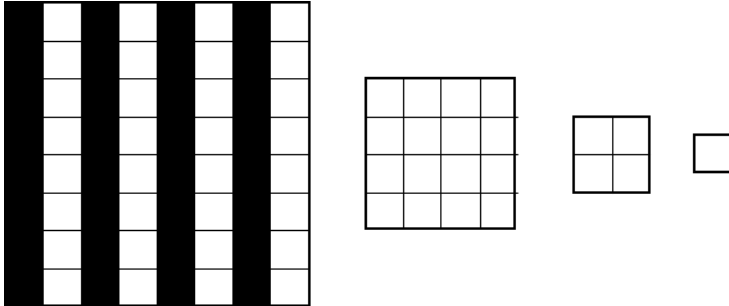
The depth range goes from n to f (near to far), and the image window (l, r, b, t) is defined on the near-plane



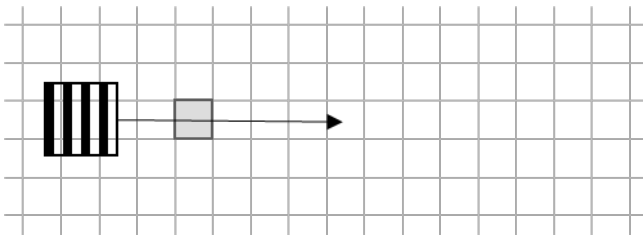
$$M_{perspective}(l, r, b, t, n, f) = \begin{pmatrix} \frac{2n}{r-l} & 0 & \frac{l+r}{r-l} & 0 \\ 0 & \frac{2n}{t-b} & \frac{b+t}{t-b} & 0 \\ 0 & 0 & -\frac{f+n}{f-n} & -\frac{2fn}{f-n} \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

Assignment 3: Textures [10 points]

a) **[1P]** Sketch the MIP-map of the following texture (use hatching  to mark 50% grey or write grey values, ideally as fractions like $\frac{1}{4}$ into the boxes)



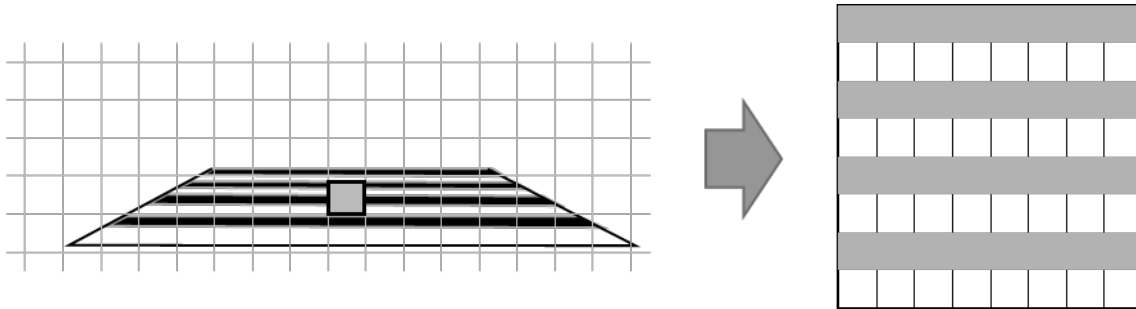
The texture is applied to a small square which moves slowly from left to right. The grey grid is the pixel grid.



b) **[1P]** Assume we rasterize without any anti-aliasing. What colors does the marked pixel get, while the square moves from left to right?

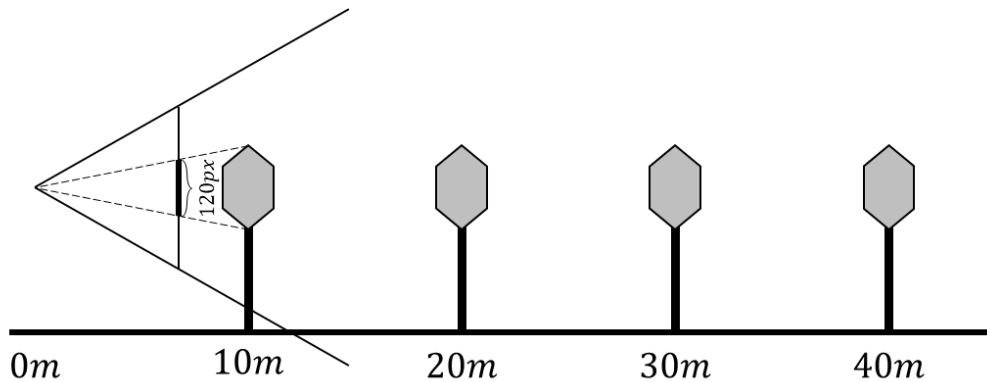
c) **[1P]** What do we see in the pixel if MIP-mapping is applied while the square moves?

b) **[2P]** The texture is now applied to a square, which is projected perspectively, as shown on the right. Sketch the footprint of the marked pixels in the texture (black pixels turned grey so you can better draw the footprint).



d) **[1P]** Why is anisotropic MIP-mapping particularly helpful in the above example? Argument with the footprint's shape.

f) **[2P]** Given is the following scene with four equivalent traffic signs within the depicted view frustum. In the perspective image, the leftmost traffic sign (in $10m$ distance to the camera) is 120 pixels tall (only the sign, without the post).



The other traffic signs in the image are of the same height in world space and located at $20m$, $30m$ and $40m$, respectively. How tall are those in image space?

g) **[2P]** A texture is mapped to the traffic signs. For the first sign at $10m$, the level of detail algorithm decides that MIP map level 8 is exactly optimal. For each of the other three traffic signs, state which level(s) will be used in trilinear interpolation (smaller level means coarser texture)!

Assignment 4: Polygon Rasterization [9 points]

a) **[5P]** The following image shows an Edge Table for a polygon. Sketch the corresponding polygon into the grid below the Edge Table! You do not have to fill single pixels, just sketch the outline of the polygon!

y_{lower}	x_{lower}	y_{upper}	$1/m$	$next$
1	1	5	0	
1	1	2	5	
2	6	4	0	
4	6	7	-1	
5	1	7	1	

7								
6								
5								
4								
3								
2								
1								
0								
	0	1	2	3	4	5	6	7

b) **[2P]** How does the active edge table look like for scanline $y = 3$?

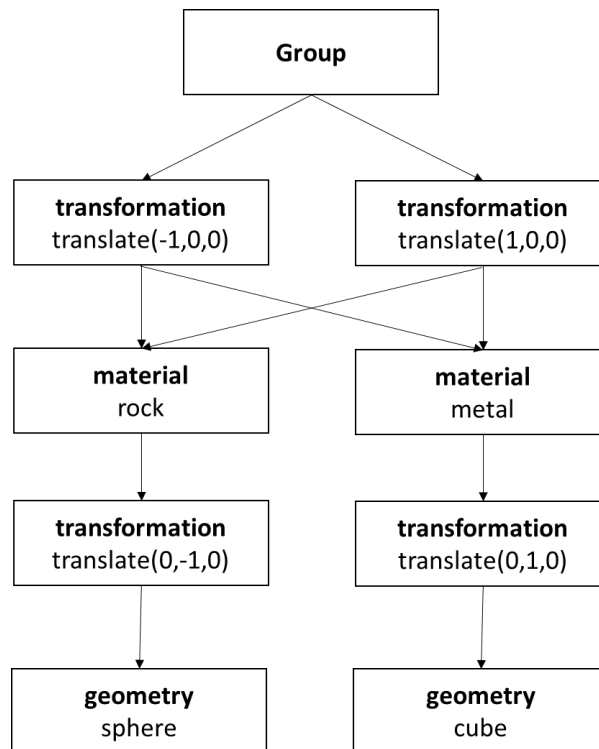
x_{inter}	$1/m$	y_{upper}	$next$

c) **[2P]** For depth tests, we also need a z-value for the set pixels. We thus provide for each vertex not only (x, y) , but also a z-value.

By which values do we have to extend the edge table and active edge table?

Assignment 5: Scene Graphs [10 points]

Given the following scene graph:



a) **[4P]** How many instances of “**geometry sphere**” and “**geometry cube**” are generated? What are their attributes? Fill in the following table with all generated instances including their geometry (sphere or cube), their material (rock or metal), and their accumulated transformation (translation vector).

Instance #	geometry (sphere or cube)	material (rock or metal)	accumulated transformation
0			
1			
2			
3			
4			
5			

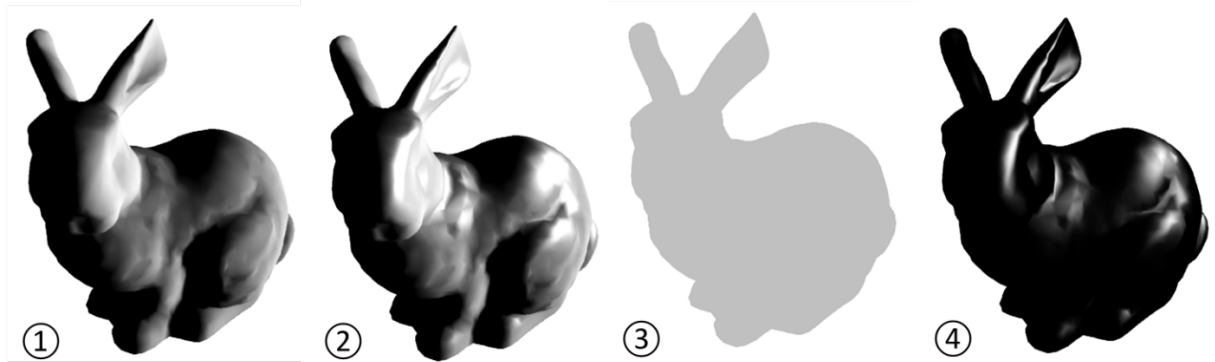
b) **[4P]** Sketch a scene graph for a car, composed of its body (**geometry** “body”) with **material** “steel” and four wheels (**geometry** “wheel”) with **material** “aluminum”. You do not need to provide attributes for **transformation** nodes. Use as few nodes as possible.

c) **[2P]** Change the scene graph so that the front wheels have the new material “wood”. Use the material nodes “steel”, “aluminum”, and “wood”, as well as the geometry nodes “wheel” and “body” only once!

Assignment 6: Phong Lighting & Shading [8 points]

Phong Lighting

a) **[2P]** The following images show a bunny, illuminated by a parallel light source. The bunny is rendered using the Phong lighting model.

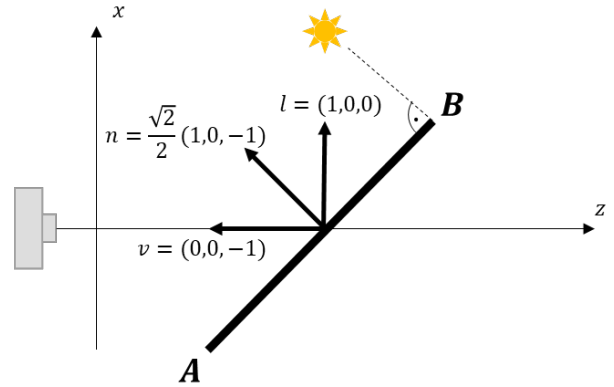


Assign the following Phong lighting parameters to the images ①, ②, ③, or ④

Ambient	Diffuse	Specular	Shininess	①, ②, ③, or ④
0.7, 0.7, 0.7	0, 0, 0	0, 0, 0	12	
0, 0, 0	0.5, 0.5, 0.5	0, 0, 0	24	
0, 0, 0	0, 0, 0	1, 1, 1	10	
0, 0, 0	0.5, 0.5, 0.5	1, 1, 1	10	

Lighting in Ray Tracing

An eye ray hits a triangle as shown in the right figure, the normal of the triangle is $\frac{\sqrt{2}}{2}(1,0,-1)$. The scene is lit by a point light in direction $(1,0,0)$, as shown on the right (incoming light $I_{in} = (1,1,1)$, no attenuation). You can approximate $\frac{1}{2}\sqrt{2} \approx 0.7$



b) **[2P]** Compute the diffuse lighting with $k_{diff} = (0.5, 0.5, 0.5)$:

c) **[2P]** Compute the specular lighting with $k_{spec} = (1, 1, 1)$ and exponent $n_{shiny} = 20$:

d) **[2P]** Sketch the diffuse lighting (left) and specular lighting (right) over the patch AB from A to B (assuming ray tracing or Phong Shading, **not** Gouraud Shading)

diffuse lighting	specular lighting

